UTS 36104 – Data Visualisation and Narratives

Assignment 3, Part 2 – Group 10 Submission

Project: Real Wages, Real Lives – Australia's Cost-of-Living Story (Q4 2023 → Q4 2025)
Marker Guide – Step-by-Step Walkthrough

How to use this guide

This document is a **step-by-step path** for the marker. Follow steps 1–8 in order; each step points you at a specific deliverable, tells you what to look at, and notes which rubric criterion it evidences. The full data dictionary and post-feedback iteration log sit at the end for reference.

Estimated total marking time: **20–25 minutes** (3-min video + 12–15 min app tour + 5 min code skim).

1. Where the deliverables live

- **Code (zip)** – attached to this submission. Full repository: `app.py`, `data/`, `requirements.txt`, `README.md`.
- **GitHub repository** – `https://github.com/ngolethebach/DataVisual_Asm3`
- **Live Streamlit web app** – `https://group10datavisualisation.streamlit.app/`
- **3-minute video walkthrough** – attached as `Video_Walkthrough.mov`. Narrated tour of the four advanced features (context-aware filtering, visual tooltips, narrative scrollytelling, what-if parameterisation) *plus* the post-feedback LGA layer.
- **This PDF** – marker guide. **Data dictionary** reproduced in §4; also documented in the project `README.md` (§4).

2. Step-by-step walkthrough

Step 1 – Open the live app (30 sec)

Open `https://group10datavisualisation.streamlit.app/` in a recent Chrome/Edge/Firefox. The page loads top-to-bottom as five chapters following the *What → So What → What Next* executive-efficiency arc. *If the app is cold-starting, give it ∼15 seconds.*

Step 2 – Watch the 3-minute video walkthrough (3 min)

Open `Video_Walkthrough.mov`. The narration directs your attention to each advanced feature in turn and frames it against the rubric (technical prowess, accuracy, relevance). *Watch before the app tour so the in-page interactions land in context.*

Step 3 – Chapter 1: "The Headline Number Lies" (3 min)

What to do. Move the **spotlight-quarter slider** in the left sidebar. The three KPI tiles (WPI, CPI, real-wage delta), the national timeline highlight, and the per-quarter callout all update in lock-step.

What it evidences.

- **Advanced feature 1 – Context-aware filtering.** One control drives multiple, semantically linked views simultaneously.
- **Narrative arc – "What".** Wage growth $\neq$ real wage growth; the gap has a recognisable shape across 2023–2025.

Where in code: `app.py` – `section_what_national()`. The slider scope is intentionally narrowed to Chapters 1–2 (see §3, item 3).

Step 4 – Chapter 2: "Eight Australias, Eight Pay Stories" + the LGA picture (4 min)

What to do.

1. Hover any state on the **geo-bubble map** – a rich tooltip surfaces WPI, CPI, real wage growth, and the rank against the national average.
2. Scroll to the **Below the State Line: The LGA Picture** subsection (*added in response to marker feedback – see §3, item 1*). Use the **Jurisdiction** radio to filter (e.g. select NSW); the choropleth, the *population by IRSD decile* bar, the most/least-disadvantaged extremes tables, and the narrative pull-quote all update from the live dataframe.
3. Toggle **View mode** between "National decile (banded)" and "IRSD score (continuous)" to see two encoding strategies on the same geometry.

What it evidences.

- **Advanced feature 2 – Visual tooltips.** Tooltips are designed, not default Plotly — units, formatting, and signposting are explicit.
- **Spatial dimension at two grains.** State-level WPI and sub-state LGA SEIFA, both cartographic; the LGA layer reveals dispersion an 8-state average necessarily hides.
- **Dataset-driven narrative.** The pull-quote numbers (population share in bottom 3 deciles, IRSD spread) are computed from the filtered view — not hard-coded.

Where in code: `section_what_state()` (state map) and `_render_lga_layer()` (LGA layer, called at the end of `section_what_state`).

Step 5 – Chapter 3: "Where the Money Actually Went" (2 min)

What to do. Toggle the three category checkboxes (**Housing**, **Food**, **Transport**) above the chart. Each adds a trace to the cumulative CPI sub-category chart against the All Groups average. The pull-quote underneath cites the cumulative gap each category has opened against the headline.

What it evidences.

- **Advanced feature 3 – Narrative scrollytelling.** The chapter is a guided argument (housing CPI is the dominant driver), with the chart a supporting witness rather than a dashboard tile.

Where in code: `section_so_what_drivers()`.

Step 6 – Chapter 4: "When Do Real Wages Recover?" (3 min)

What to do. Use the three sliders (forecast horizon, projected WPI %, projected CPI %) to parameterise the recovery scenario. The chart projects a cumulative-real-wage line forward; the verdict text underneath is generated dynamically based on whether the recovery threshold is crossed within the horizon.

What it evidences.

- **Advanced feature 4 – What-if parameterisation.** The reader sets the conditions; the chart and the verbal verdict both respond.

Where in code: `section_what_next_whatif()`.

Step 7 – Chapter 5: Findings & Next Steps (2 min)

What to do. Read the five recommendation cards. Each pairs a number from Chapters 1–4 with a Treasury-framed next step. Note that the numbers are computed at render time from the loaded dataframe (*see §3, item 2*).

What it evidences.

- **Communicating to stakeholders.** The recommendations land as analyst briefing points, not as generic "insights". Each is dataset-anchored and traceable.

Where in code: `section_call_to_action()`.

Step 8 – Skim the source (5 min, optional)

Open `app.py` in the attached zip (or on GitHub). It is a single annotated module: a top docstring documents the narrative arc and the four advanced features; section banners separate constants, layout/CSS, and chapter functions; inline comments explain non-obvious choices. `README.md` mirrors this structure for orientation without reading the code itself.

3. Post-marker-feedback iterations

Three items were raised in marker feedback on the draft and have been addressed in this final submission:

1. **Sub-state (LGA) visualisation added.** A new IRSD-by-LGA SEIFA layer is wired into Chapter 2 (565 LGAs, ABS 2011 boundaries), with jurisdiction filter, view-mode toggle, population-by-decile bar, and most/least-disadvantaged extremes tables. See Step 4 above.
2. **Recommendations are now dataset-driven.** The five Chapter 5 cards previously hard-coded their headline numbers; they now read from the loaded dataframe so any regenerated CSV automatically updates the recommendation copy.
3. **Spotlight-quarter slider re-scoped.** Previously visible globally and applied to every chapter (including the cumulative-drivers chapter, where a single-quarter snapshot was misleading). The slider is now scoped to Chapters 1–2, with an on-page scope note explaining the boundary.

4. Data Dictionary

Quarterly dataset – `data/wage_inflation.csv`

Shape: 72 rows × 15 columns **Coverage:** 2023 Q4 → 2025 Q4

Column	Type	Description	Source
`quarter_date`	date (ISO 8601)	First day of the quarter, e.g. `2024-03-01`. Use as the time axis.	Derived
`quarter`	str	Human-readable label, e.g. `2024 Q1`. Good for chart tick labels.	Derived
`year`	int	Calendar year (2023–2025).	Derived

Column	Type	Description	Source
`quarter_num`	int	ABS quarter convention: Q1 = March, Q2 = June, Q3 = Sep, Q4 = Dec.	Derived
`state_code`	str	Short code: NSW, VIC, QLD, SA, WA, TAS, NT, ACT.	Constant
`state_name`	str	Full state/territory name.	Constant
`latitude`	float	Capital-city latitude for mapping (e.g. Sydney = −33.87 for NSW).	Constant
`longitude`	float	Capital-city longitude.	Constant
`wage_index`	float	**Wage Price Index** for that state – Original series, total hourly rates excl. bonuses, Private + Public, all industries. Re-based Sep 2008 = 100. **The only column that varies by state.**	634502b.xlsx
`inflation_rate`	float	National CPI **All Groups** QoQ % change, weighted 8-capital-city average.	6401018.xlsx
`wage_growth`	float	National WPI QoQ % change, Private + Public, all industries – Original (not seasonally adjusted).	634501.xlsx
`real_wage_growth`	float	`wage_growth` − `inflation_rate`. Positive = real purchasing power increasing; negative = declining.	Computed
`food`	float	National CPI **Food and non-alcoholic beverages** QoQ % change.	6401018.xlsx
`housing`	float	National CPI **Housing** QoQ % change.	6401018.xlsx
`transport`	float	National CPI **Transport** QoQ % change.	6401018.xlsx
`cum_real_wage`	float	Running cumulative sum of `real_wage_growth` from series start. Added at load time.	Computed in `app.py`
`real_sign`	str	`"positive"` / `"negative"` / `"zero"` – used for conditional bar colouring.	Computed in `app.py`

Note on `wage_growth` (Original vs Seasonally Adjusted). The Original ABS series is used because it matches the headline numbers in ABS media releases and is easier to explain to a general audience. Seasonal noise – a Q3 spike due to annual award-rate decisions – is real and expected, not a data error. If a smoother trend line is needed, swap to the Seasonally Adjusted series from `634501.xlsx`.

Sub-state LGA dataset – `data/lga_boundaries_2011.json` + `data/lga_seifa_2011.json`

Matched-pair dataset added in response to marker feedback. Both files originate from ABS Census 2011, key on `LGA_CODE11` / `lga_id`, and merge 1:1 across 565 LGAs (one geo-only "no usual address" placeholder is intentionally unmatched).

File	Records	Contents
`lga_boundaries_2011.json`	565 GeoJSON features	Polygon geometry per LGA + properties `STATE_CODE`, `LGA_CODE11`, `LGA_NAME11`. ABS 2011 ASGS boundaries. Drives the Chapter 2 LGA choropleth.

File	Records	Contents
`lga_seifa_2011.json`	565 records	SEIFA **IRSD** (Index of Relative Socio-Economic Disadvantage) by LGA. Fields: `lga_id`, `lga_name`, `state`, `score`, `national_decile`, `national_percentile`, `state_decile`, `state_percentile`, `state_rank`, `national_rank`, `population`. Lower IRSD score = more disadvantaged.

Why 2011 vintage? The two files are a matched pair from a single Census; using a newer 2021 SEIFA against 2011 boundaries (or vice versa) would break the join because of post-2016 NSW LGA amalgamations. The IRSD is treated as a *structural* indicator – the relative ranking of LGAs is comparatively stable across vintages, even if the absolute score moves.

Joining strategy – the enrichment layer

The key insight driving the dataset design: **WPI is published per state, but CPI sub-categories are only available at the national level.**

1. **WPI per state** pulled from `634502b.xlsx` – spatial dimension.
2. **National CPI metrics** (All Groups, Food, Housing, Transport) pulled from `6401018.xlsx` – single national series per quarter.
3. **National WPI QoQ %** pulled from `634501.xlsx`.
4. **Broadcast** national CPI/WPI values across all 8 state rows for each quarter (long-format pattern). Values repeat per state within a quarter – this is correct and expected.
5. **Capital-city lat/long** attached to each state so map libraries work without additional joins.
6. **9 most recent quarters** selected (2023 Q4 → 2025 Q4) for a clean 72-row dataset with sufficient trend length.
7. **LGA layer** loaded separately at app start; not joined to the quarterly frame because SEIFA is a structural Census snapshot, not a quarterly series.
